



Experiment no:-1

Objective: - Installing oracle/ MYSQL Procedure

Download My SQL Installer

If you want to install MySQL on Windows environment, using MySQL installer is the easiest way. MySQL installer provides you with an easy-to-use wizard that helps you to install MySQL with the following components:

- My SQL Server
- All Available Connectors
- MySQL Workbench with Sample Data Models
- My SQL Notifier
- Tools for Excel and Microsoft Visual Studio
- MySQL Sample Databases
- MySQL Documentation

To download MySQL installer.

Install MySQL via MySQL Installer

To install MySQL using the MySQL installer, double-click on the MySQL installer file and follow the steps below.

Step1: Windows configures MySQL Installer

Step2 -Welcome Screen: A welcome screen provides several options. Choose the first option: Install MySQL Products

Step3- Download the latest MySQL products: MySQL installer checks and downloads the latest MySQL products including MySQL server, MySQL Workbench, etc.

Step4: Click Next button to continue

Step5- Choosing a Setup Type: there are several setup types available. Choose the Full option to install all MySQL products and features.

Step6-CheckingRequirements



Step7- Installation Progress: MySQL Installer downloads all selected products. It will take a while, depending on which products that you selected and the speed of your internet connection.

Step7-Installation Progress: downloading Products in progress.

Step7-Installation Progress: Complete Downloading. Click Next button to continue...

Step8- ConfigurationOverview.ClickNextbuttonconfigureMySQLDatabaseServer

Step8.1 - MySQL Server Configuration: choose Config Type and MySQL port (3006by default) and click Next button to continue.

Step8.1 - MySQL Server Configuration: choose a password for the root account. Please note the password download and keep it securely if you are installing MySQL database server on a production server. If you want to add a more MySQL user, you can do it in this step.

Step8.1 -MySQL Server Configuration: choose Windows service details including Windows Service Name and account type, then click Next button to continue.

Step8.1 - MySQL Server Configuration – In Progress: MySQL Installer is configuring MySQL database server. Wait until it is done and click Next button to continue.

Step8.1-MySQLServerConfiguration-Done.ClicktheNextbuttoncontinue.

Step8.2 - Configuration Overview: MySQL Installer installs sample databases and sample models.

Step9- Installation Completes: the installation completes. Click finish button to close the installation wizard and launch the MySQL Workbench.

Result:-MYSQL is installed completed.



Experiment no :- 2

Objective:-Creating Entity-Relationship Diagram using case tools.

Procedure:-

E-R model:-

The ER or (Entity Relational Model) is a high-level conceptual data model diagram. Entity-Relation model is based on the notion of real-world entities and the relationship between them.

ER modeling helps you to analyze data requirements systematically to produce a well-designed database.

Entity relationship diagram displays the relationships of entity set stored in a database. In other words, we can say that ER diagrams help you to explain the logical structure of databases. At first look, an ER diagram looks very similar to the flowchart. However, ER Diagram includes many specialized symbols, and its meanings make this model unique.

Component of ER Model

Entity relationship model having three components are

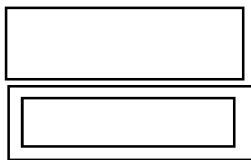
- Attributes
- Relationship

Entity:- A real-world thing either living or non-living that is easily recognizable and no recognizable. It is anything in the enterprise that is to be represented in our database. It may be a physical thing or simply a fact about the enterprise or an event that happens in the real world. An entity can be place, person, object, event or a concept, which stores data in the database. The characteristics of entities are must have an attribute, and a unique key. Every entity is made up of some 'attributes' which represent that entity.

Attribute:-Attribute are unit that describe the characteristics are properties of entity.

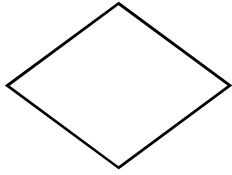
Relationship:-Relationship is nothing but an association among two or more entities.

Notations are used in ER diagram



Strong Entity

Weak Entity

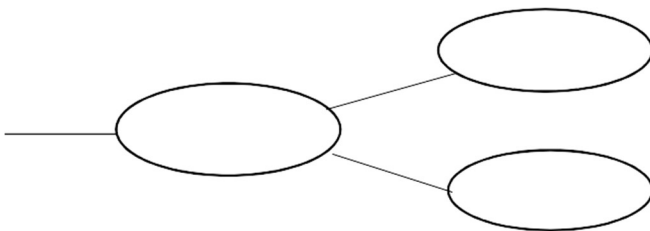


Relationship



Attribute

Attribute Key



Composite Attribute

Multivalued Attribute



Derived Attribute

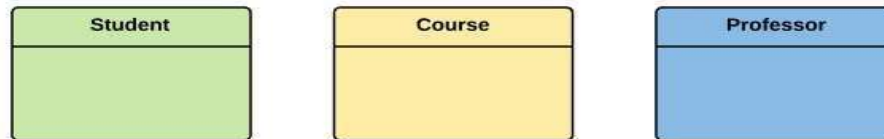
Step to Create ER diagram :-



Step1) Entity Identification

We have three entities

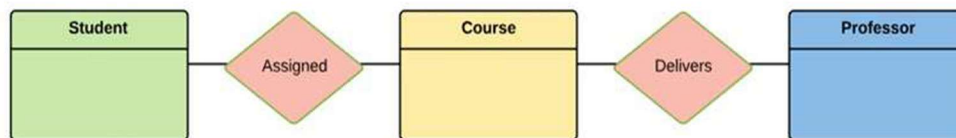
- Student
- Course
- Professor



Step2)Relationship Identification

We have the following two relationships

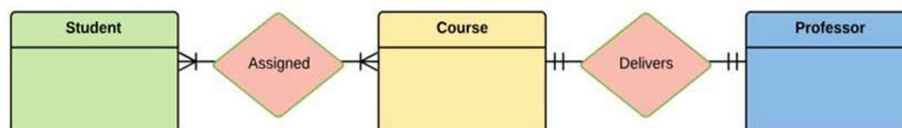
- The student is **assigned** a course
- Professor **delivers** a course



Step3)Cardinality Identification

For them problem statement we know that,

- A student can be assigned **multiple** courses
- A Professor can deliver only **one** course



Step4)Identify Attributes

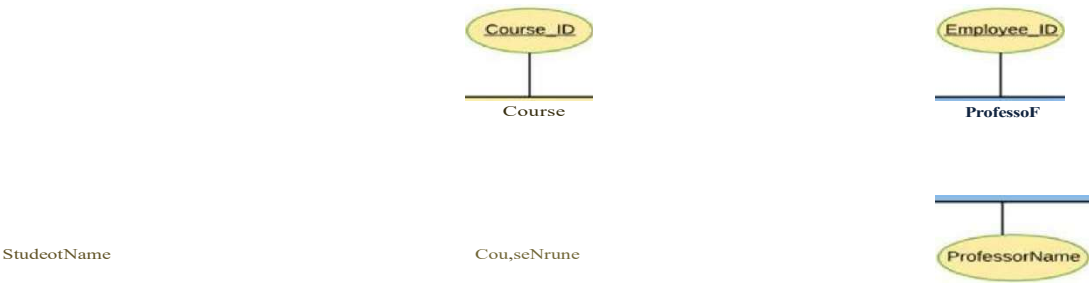
You need to study the files, forms, reports, data currently maintained by the organization to identify attributes. You can also conduct interviews with various stakeholders to identify entities. Initially, it's important to identify the attributes without mapping them to a particular entity.

Once, you have a list of Attributes, you need to map them to the identified entities. Ensure an attribute is to be paired with exactly one entity. If you think an attribute should belong to more than one entity, use a modifier to make it unique.

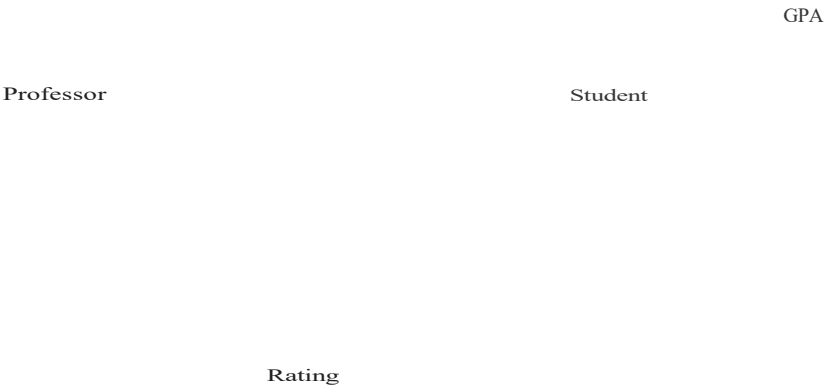
Once the mapping is done, identify the primary Keys. If a unique key is not readily available, create one.



Entity	Primary Key	Attribute
Student	Student ID	Student Name
Professor	Employee_ID	Professor Name
Course	Course ID	Course Name



Step5)Create the ERO





Experiment no :-3

Objective:- Writing SQL statements Using ORACLE/MYSQL:

- ❖ Writing basic SQL SELECT statements.
- ❖ Restricting and sorting data.
- ❖ Displaying data from multiple tables.

Procedure:-

■ SQL SELECT statements Syntax

```
SELECT column1, column2...column  
N FROM  
        tablename;
```

Example:-

Table-members

membership_ number	Full names	gender	Date of birth--	physical_ address	postal_ address	contact_ number
1	Janet Jones	FemaIe	21-07-1980	First Street Plot No4	P. t e d a g n v a	0759253 542

The basic SQI SELECT statement are

```
SELECT'full_names','gender','physical_address','email'FROM'members';
```

■ Restricting and sorting data Syntax

```
SELECT*!{[DISTINCT]column ! expression[alias], ... }  
FROM table  
[WHERE condition (s)]
```

For example :-

SELECT query lists the first names of those employees whose last name starts with 'SA'.

```
SELECT first_name
```

```
FROM employees
```

```
WHERE last_name 'SA%';
```

■ Displaying data from multiple tables Syntax

```
SELECT Orders. OrderID, Customers.CustomerName, Orders.OrderDate  
FROM Orders  
INNERJOIN Customers  
ONOrders.CustomerID=Customers.CustomerID;
```

■ Aggregating data using group function syntax

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- AVG, that calculates the average of the specified columns in a set of rows,
- COUNT, calculating the number of rows in a set.
- MAX, calculating the maximum,
- MIN, calculating the minimum,
- STDDEV, calculating the standard deviation,
- SUM, calculating the sum,
- VARIANCE, calculating the variance.

The general syntax for using Group functions is:

```
SELECT <column, ...>,group_function(column)
```

```
FROM <table>
```

```
WHERE <condition>
```

```
[GROUP BY <column>]
```

```
[ORDERBY<column>]
```

Note that the column on which the group function is applied must exist in the SELECT column list.

For example, the following selects the maximum salary from the employee table.

```
SQL> SELECT dept, MAX(sal)
```

```
FROM employee;
```

Using Group functions

You can use AVG and SUM for numeric data. MIN and MAX can be applied on any datatype including numbers, dates and strings. The order is defined accordingly.

For example, the following statement forces to take the average for all salary values even including 0 (although it is an odd case, let's assume this is possible.)

```
SQL>SELECT dept, AVG(NVL(sal,0))
```

```
FROM employee;
```

Using Group-By Clause

NotethatALLcolumnsintheSELECTlistthatarenotingroupfunctionsmustbe in the GROUP-By clause.

For example,



```
SQL> SELECT dept,job,  
        MAX(sal) FROM employee  
        GROUPBY dept,job;
```

Using Having-Clause

To exclude some group results, use Having-clause.

For example,

```
SQL>SELECT dept,MAX(sal)  
        FROM employee  
        GROUPBY dept  
        HAVINGMAX(sal) >2000;
```

Note that you CANNOT use WHERE clause to restrict groups. For example, the following is illegal:

```
SQL>SELECT dept, MAX(sal)  
        FROM employee  
        WHEREMAX(sal)>2000  
        GROUPBY dept;
```

This is because the group function cannot be used in WHERE clause.

Nesting Group Functions

You can nest group function calls. For example, the following example to show the maximum of the average salary of all departments,

```
SQL>SELECTMAX(AVE(sal))  
        FROM emp  
        GROUPBY dept_no;
```